

29,000 transistors
 5, 8, 10 MHz
 40 Pins -
 16 bit Processor (16 bit data line).
 8080/80180 have internal architecture of 16 bit but their data line are only 8.
 - 16 bit, N channel HMOS II P (High performance MOSFET).
 27 / 21 8 pins dual ↑ I/O Pins (bidirectional).

21 → Reset
 22 - Ready
 23 - TEST
 24 - $\overline{QS1}$ (INTA)
 25 - ALE ($\overline{QS0}$)
 - 27 - ($\overline{MIO}$)
 A16 - A19 -
 0 to 15 (AD0 to AD15)
 16 bit Multiplexed.

NMI, $\overline{BHE}$ /
 20 address lines | $2^{20} (K \times K) = 1 \text{ Mbytes}$.
 16 data lines.
 2 ground pins (2 pins connected to ground).
 AD0 - AD15 - works in 2 Modes

Ready - Active high signal, peripheral is ready to transfer data.
 Active low O/P signal used for read operation.
 $\overline{RD}$ - HP to memory / IO device etc.

AD0 to AD15 Bidirectional
 16 bit address / Data lines.

A16 to A19 - Higher order Address Bus.
 20 Address lines ✓
 $\overline{BHE} / S7$ - Bus high enable / Status

$\overline{MN} / \overline{MX}$ → Processor in Minimum or Max Mode
 0 - max Mode
 1 - min mode.
 NMI - Non Maskable interrupt

Request
 $\overline{TEST}$ → Active low signal.
 (low → HP will execute Normally)
 (high → HP will wait)

S1 - S6 → status signal
 Multiplexed with higher order address lines.

Mode 0 →
 $\overline{MIN}$ - $\overline{MN}$

MN/Min Mode: in this single HP operates in min mode. ①
 - In this processing unit issues Control signal required by memory and I/O devices.

MAX MODE :-

Bus Controller issues control signals.

Bus Controller $\rightarrow$ Control signal.

Processor or Bus Controller

who is delivering signal.

Max Mode

Min Mode

- | | |
|----|--|
| 24 | $\overline{INTA}$ - Interrupt Ack. |
| 25 | ALE - Address latch en |
| 26 | $\overline{DEN}$ - O/P data enable. |
| 27 | $DT/\overline{R}$ - Data Transmit/Receive. |
| 28 | $M/\overline{IO}$ - Mem/I/O Access. |
| 29 | $\overline{WR}$ Write |
| 30 | $H/LEDA$ - Hold Ack |
| 31 | $HOLD$ - Hold Request. |

$QS1$
 $QS0$ } Queue Status

$\overline{S0}$
 $\overline{S1}$
 $\overline{S2}$ } Status signal.

$\overline{LOCK}$ - locks the peripheral.

$\overline{PQ} / \overline{GT}_1$ - local bus priority control.

$\overline{RQ} / \overline{GT}_2$ - local bus priority.

Processor will deliver signals.

Diff signals.

Architecture

Bus interface $\rightarrow$ Data, Address, Control etc

Execution unit $\rightarrow$ To execute program

- segments register, instruction on and 6 byte instructions

are part of (BIU)

- general Purpose Registers, Pointer, BP, instruction Reg,

16 bit data bus + 16 bit architecture

Nchannel HMOS (518, 10) MHz

27 O/P, 21 input Pins (8 pins), are bidirectional

AD₀ - AD₁₅ (Pin 16 to Pin 2), Pin 1 = ground

2 to 16 - 16 Pins (AD₀ - AD₁₅)

16 bit of Address + data

are multiplexed

AD₀ to AD₁₅

Pin 2 to Pin 16 } 16 multiplexed Address + Data Pins

8086 used 20 address lines and 16 data lines

2^{20} (locations can be addressed)

2 MB 1 megabyte

AD₀ - AD₁₅ - Can work on Address / Data Mode

2 Pins for ground -

V_{CC} - 1 Pin

Reset - To Reset Complete Processor (All Reg's Value will be 0)

Bus becomes ∞ (high impedance)

$\overline{INTA}$ is just for $\overline{INTR}$

Ready: Active high input signal indicating peripherals ready to transfer data.

$\overline{RD}$ = For Read operation HP to mem / I/O device

AD₀ - AD₁₅ : Bidirectional latched 16 lines

A₁₆ - A₁₉ - Higher order address bus

$\overline{BNE} / S_7$: Bus high enable / status

$\overline{MN} / \overline{MX}$ $\Rightarrow$ HP operating in Minimum / Max Mode

$\overline{NMI}$ - Non Maskable interrupt Request

$\overline{TFSI}$ - Active low signal - (low $\rightarrow$ μP continues execution, and when it's high μP goes to wait state)

S1-S6 → Status signal Multiplexed with higher order address lines. ①

2 modes it can operate:

① mn/min Mode - in this Processing unit issues control signals, Required by memory & I/O (single processor) system

② mx/max Mode - In multiprocessor system, 8086 operates in max

Mode - In this bus controller issues control signals,

who delivers control signals — ① Processing Unit
or ② Bus Controlling unit

min Mode & max Mode signals

min Mode

24: $\overline{INTA}$ interrupt Acknowledge
 25: $\overline{ALE}$ Address latch enable
 26: $\overline{DEN}$ - O/P data enable
 27: $\overline{DT/R}$ Data transmit/receive
 28: $\overline{M/\overline{IO}}$ Mem/I/O
 29: $\overline{WR}$
 30: $\overline{HLD A}$ - Hold Ack now
 31: $\overline{HLD P}$ - Hold Program system bus.
 31: Hold bus.

Processor sends Control sig.

Max Mode

$\overline{QS1}$ - Queue status
 $\overline{QS0}$ -
 $\overline{S0}$ } Status signal
 $\overline{S1}$ }
 $\overline{S2}$ }
 $\overline{LOCK}$ - Lock peripheral
 $\overline{RQ}/\overline{GT1}$ } local Bus priority
 $\overline{RQ}/\overline{GT1}$ } Control -
 Bus Controller sends
Control Signals

divided into 2 fundamental, functional unit

① Bus interface unit [BIU] - Data bus / Address / Control Bus. ①

② Execution unit [EU] - To execute Program ②

6 byte instruction? —

EU

execution unit

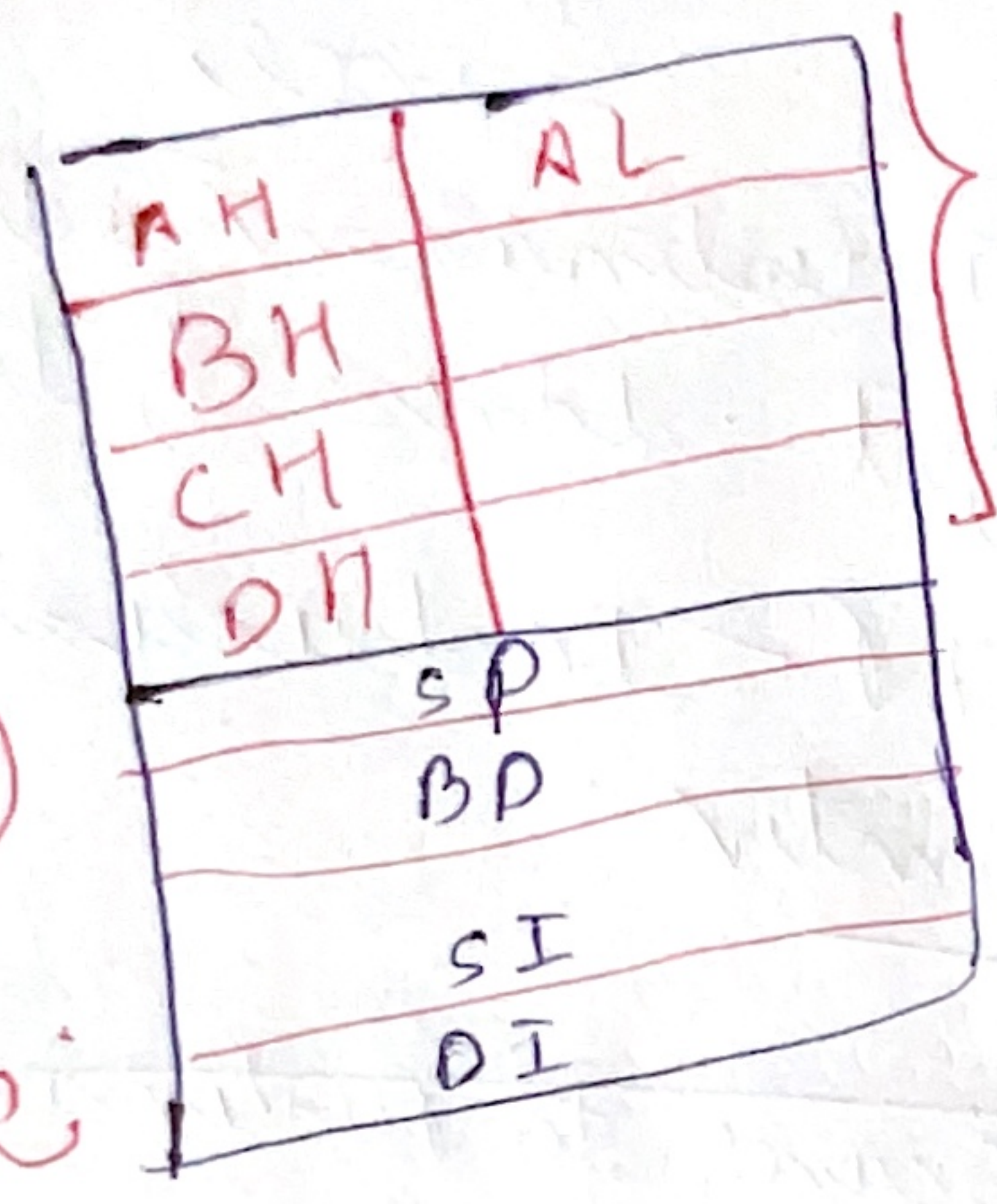
is a Part of Bus interface unit.

① general Purpose Registers
 ② Pointer
 ③ base Pointer
 ④ ALU instruction, decoder are part of this

ALU flags, instruction decoder, Timing & Control bits,

Register
Pointer
Base
Instruction decoder.

} CPU



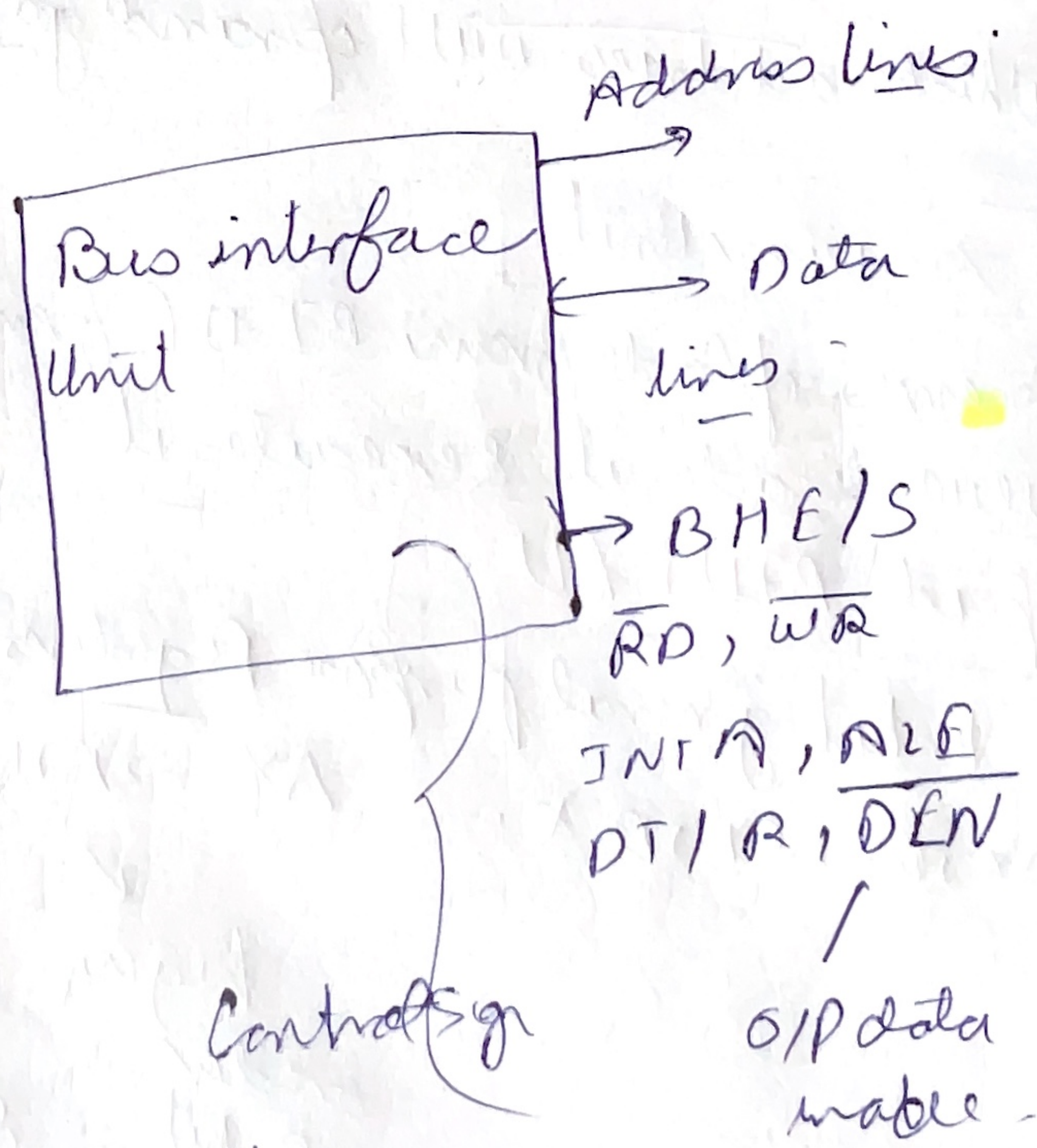
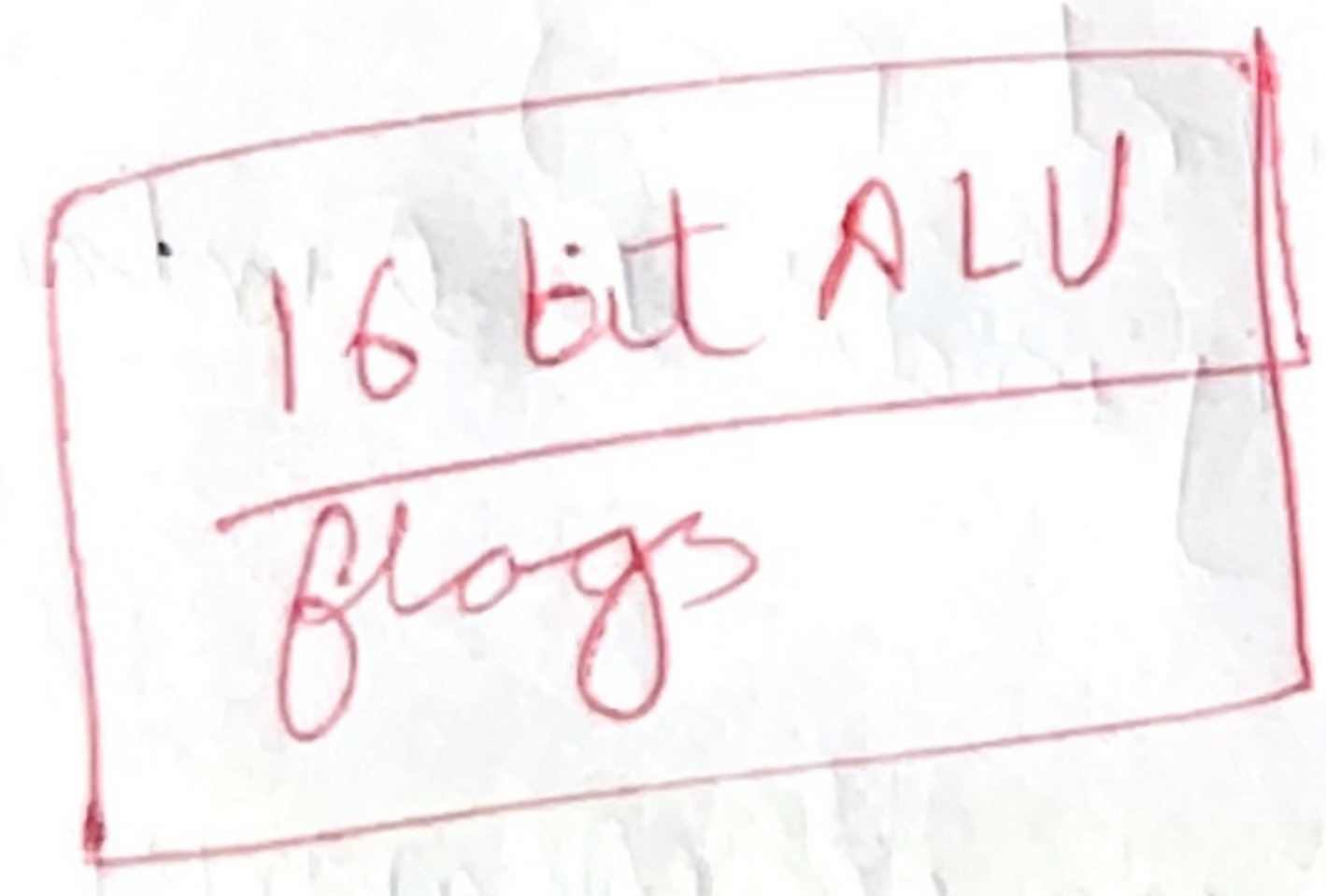
general Purpose

Index registers



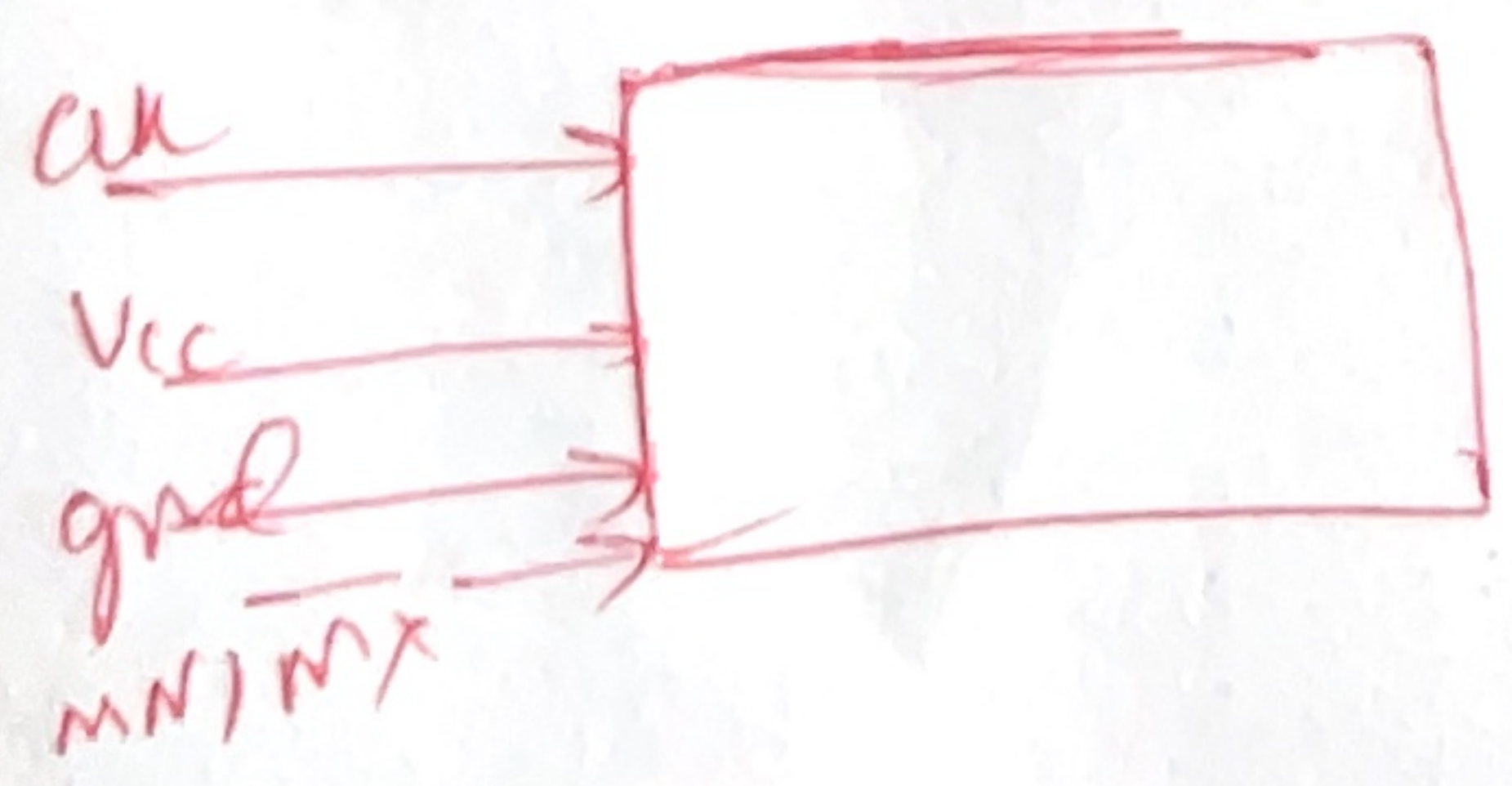
segment registers

Instruction Pointer



Control Sign

6 byte instruction queue
(line of instructions)



BIU & EU Bus interface Unit
Works parallelly Bus interface
Execution Unit

In branching.

- We need a new instruction queue. (6 inst)
- We need to dump instructions from a new location.

Overlapped execution (6 instructions)

Reads { BIU (Bus interface unit)
Write }

Relocate addresses of operands
since it gets unrelocated operand
address from EU.

Which address will operand go to.

Execution Unit

Receives op code from BIU (from the
queue decodes it & executes it)

Speed BIU → EU

4 16 bit general purpose Registers

AX, BX, CX, DX,

AX → AH AL

highly lower
order order
8-bit 8-bit

→ AX also serves as accumulator

→ BX also serves as a base register

- CX - Purpose of Counter

- DX - also used when data is transferred

b/w I/O Port & Memory using
certain I/O instruction

Pointers and Index
Registers.

① SP (Stack pointer)

② BP (Base)

③ SI Source Index
(offset address of
Source).

④ DI destination Index
offset address of destination

⑤ IP - Instruction
Pointer

from Source to Destination

③ Segment Registers

① Code Segment (CS)

② Data Segment (DS)

③ Stack Segment

④ Extra segment (ES).

System Memory is divided into
4 segments,

Starting address of memory
segment, A segment register
points to.

64KB (Max size)

Starting Address

memory segments

RIM - in POBS To
Read interrupt status

overflow in 8005.

2000H

Code Segment

5 per digits
(20 bit address)

40,000

D S

60 1000 H

SS

80,000H

ES -

6 } 9 Total — Condition
3 } status flags — Control

in 8085 - 5 condition
+ 6th (overflow).

Control $\rightarrow$ Direction	} control flag
Trap	
Interrupt	

Used in string operation

Control flag $\rightarrow$ Set based on instruction
Condition flag will be Told by HP
based on Result of ALU.

TF $\rightarrow$ Trap for single step mode
single step mode

4 3 2 1 0
20,000

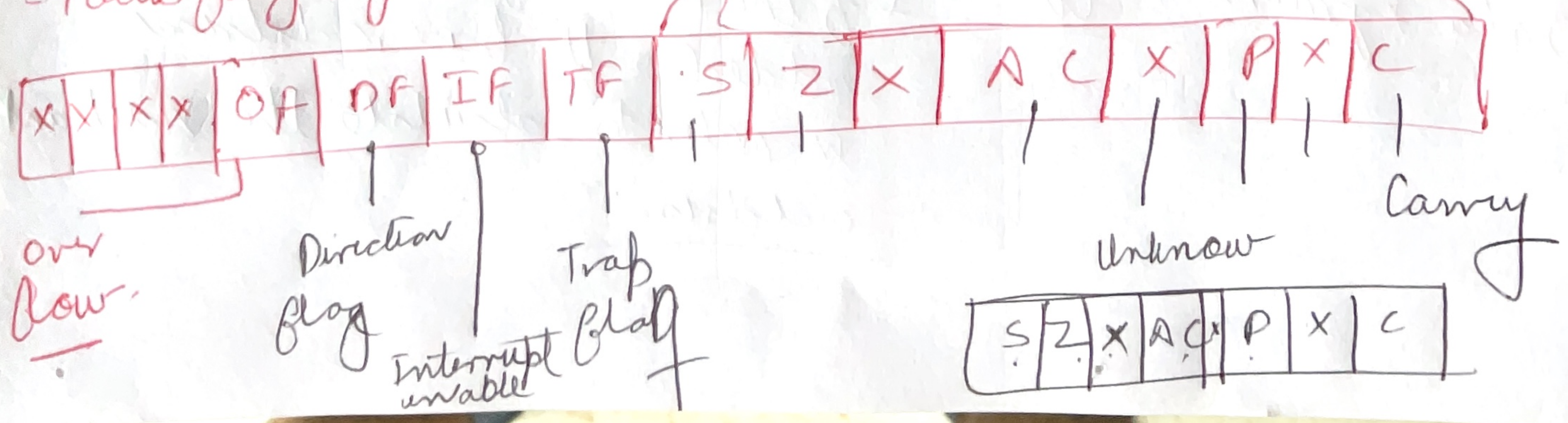
$$16^4 \times 2$$
$$= 2^{16} \times 2$$
$$2 \times 2^6 \times 2^{10}$$
$$= 2(64k)$$

IP (Instruction Pointer)

IP (Instruction Pointer)
Address of next instruction to be fetched

Status Blog Pg

Same as POBS.



Addressing Modes in 8085

MOV BX, 3598H ①

immediate data is specified in instruction itself.

② Next 6 addressing modes specify memory location of an operand

Mem Address (starting Address + offset)

Address elements

① Base Address.

BP/BX's Content.

ii) Index

it is the content of index register (SI/DI)

(iii) Displacement.

• 8/16 bit immediate value

③ Direct Addressing Mode (offset given as a 8/16 bit displace

ADD BL, [0103]

16 bit offset

• Adds content of offset address 0103H to BL

each mem segment was of 64 kB (16 address lines).

(Base 10w)

actual Address of = starting address + offset

operand

2 x 64 kB Byte (2 = 2 x 8 = 16 bit data)

data segment + offset

So here I told address inside of the Data segment, which got added to the base address in order to calculate the resultant address.

Register Addressing Mode.

8/16 bit general purpose registers

MOV AX, CX;

ADD AL, BL etc - (A, B, C, D)

are general purpose registers

Indirect Addressing Mode

MOV BX, [BP]

; Data which is in
; Base Pointer Address
; Address at this location

a hardware interrupt

NMI - Can't be disabled by software

used for emergency

② INTR -
AP will send INTA

also a programmer can
also interrupt IP by issuing
INTR instruction
software interrupts

256 h/w or software interrupt

h/w, software / internal

error

Interrupt vectors -

Those interrupts, whose

addresses (instructions) is
written at a particular
loc in memory.

Interrupt ^{INTR}
Trap ^{INT2}
m 851 → 511W - RST 4.5, 5.5, 6.5, 7.5
→ 8 software interrupt 0 to 7 RST

in 8086 - 2 h/w interrupt

① NMI (Non Maskable)

② INTR (Maskable)

Register Relative Addressing

Register + Displacement

effective address = BX + displacement

MOV CL, [BX + 05H]

⑥ Indexed Addressing

The offset is index

operand's offset is the sum of 8-bit / 16-bit displacement and the content of index register SI / DI

MOV BX, [SI + 1520H]

⑦ Based index Address

Based Register + indexed Register

offset = [BX or BP], [SI or DI]

eg ADD CX, [BX + SI]

x(Interrupt Vector - Address of ISR
Table located in first 1024 bytes

Mem

1.e 0000H To 003FFH

0000H	00003H	Type 0 - Div by 0 error
4	7	1 - Single Step
8	0000BH	Type 2 - Non Maskable.
0000C		

256 x 4 B
↑
To store vector
Address of ISR
for each interrupt

MOV Destination, source
source - Register, Mem location, immediate data
Destination → Register, Mem location

MOV CX, 037AH

MOV BX, 0301H

MOV 037AH into CX

Mem data at offset address at
0301H to BX.

Push Source

PUSH DS, POP [5000H]

↓ offset address

Pop from this offset address to
a destination

8
XCHG Destination, Source.

XCHG [5000H], AX

→ I can be a mem location.

IN AL/AX, 8 bit / 16 bit Port Address.

Reads data from specified Port Address.

DX is only Register to carry Port address.

eg IN AX, DX } Port Address in
DX.

OUT AL/AX, 8-bit / 16 bit Port Address

Writes Data at specified Port address.

OUT 50H, AL

16 bit data.

16 bit address

but addressing can be till 20 bits.

offset = address inside a Register,
That address will be searched
in address segment.

MUL →

Multiplies, Content
of Source and Reg.

Both unsigned.

MUL CL (Multiplies
AL & CL).

DIV — Divides,

Content of AL/AX

[] offset

Register

SI, DI

for string
offset

Arithmetic

ADD destination, source

ADC " " " " // Add contents of
// Carry flag also

Multiplication — AX, DX (Arithmetic Regs)
└ Accumulator, Data.

Division — quotient in AL,

Remainder in AH, for 16 bit operation

Multi processing

9

$\overline{Pn}$ connected To $\overline{gn}$ 15, $\overline{mn}/\overline{mx}$

Q 8008: $MOV [1234H], AX$

More control of AX , offset To PC
offset $[1234]$.

Q4. CX Register as a Counter -

(iii) loop + rotate -

Q 2 MB Memory - How Many Address lines

- $2 \times 2^{20} = 2^1$ address lines

$$2^{10} = k$$

$$2^{20} = M$$

8086 has an, 20 bit address bus.

→ 16 bit data bus

Lecture 10